

Total No. of Questions : 8]

SEAT No. :

P7604

[Total No. of Pages : 3

[6180]-123

T.E. (Electronics and Computer Engineering)

ADVANCED JAVA PROGRAMMING

(2019 Pattern) (Semester-I) (310342)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Explain the following methods for an applet **[6]**

- i) init ()
- ii) destroy ()
- iii) Paint ()

b) Explain the following AWT/swing components **[6]**

- i) JTextField
- ii) JFrame
- iii) JSplitPane

c) Write a program to create an applet with some background color and foreground color with a message. The message string is stored in msg and is to be displayed in paint() method. **[6]**

OR

Q2) a) Explain the following. **[6]**

- i) adapter classes
- ii) inner classes

b) Write a program to pass name and salary to an applet and get the tax calculated. Make use of get Parameter() method. **[6]**

c) Write a program to create a combo box with names of some countries. The user can select any one of them from the list and the selected country name is displayed again in the frame. **[6]**

P.T.O.

- Q3) a)** What are AWT components? Explain some AWT components. **[6]**
- b) Write a program to trap a key which is pressed on the keyboard and display its name in the TextField. (Consider F1, F2, F3 keys). **[6]**
- c) Write a Java program to create a text area and display the mouse event when the button on the mouse is clicked. **[6]**

OR

- Q4) a)** Write a short note on. **[6]**
- i) Event Classes
- ii) Event Listeners
- b) Write a program which first creates a frame and then closes it on clicking the close button. **[6]**
- c) Write a Java program that allows you to fill the shapes with some colors. **[6]**

- Q5) a)** Write short notes on the following. **[8]**
- i) Border Layout
- ii) Grid Layout
- b) Write a Java program to create a card layout and display 3 buttons, Button1, Button2 & Button3 in card Layout. **[8]**

OR

- Q6) a)** Write a Graphical User Interface program in Java to create a choice menu with names of some languages from where the user has to select any one item. The selected item must also be displayed in the frame. **[8]**
- b) Explain in detail java utilities package. **[8]**

- Q7) a)** Explain the following in JDBC **[10]**
- i) DSN
- ii) ScrollableResultSet
- iii) BLOB
- iv) PreparedStatement
- v) ODBC

- b) Explain the following in relation to JDBC [8]
- i) Class. for Name (dname) method
 - ii) Driver Manager.getConnection
(“jdbc:oracle:thin:@localhost:1521:xe”, “scott”, “tiger”);

OR

- Q8)** a) Write a JDBC program to retrieve data from the table emptab of test database. [8]
- b) Explain in brief the following. [10]
- i) ResultSetMetaData
 - ii) Reader
 - iii) Type 4 (JDBC driver)
 - iv) execute () Method
 - v) getProperty ()



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3) Assume Suitable data, if necessary.

- Q1)** a) Define Applet. Explain the types of Applet with suitable diagram in detail. **[6]**
- b) Describe the various sections of Web page in the context of Applet. **[6]**
- c) Compose an Applet program to change the Foreground and Background Color. **[5]**

Q2) a) Explain with suitable example to pass parameters to an Applet using `<PARAM>` tag. **[6]**

- b) Explain the following Swing components in detail: [6]
- i) JButton ii) Comboboxes.

- c) Compose a Java Applet program to implement a JCheck Box class. [5]

- Q3) a)** Describe the following: [6]

- i) Foreground Events and Background Events.
- ii) Event Classes.

- b) Sketch and explain the hierarchy of classes in the java. awt package.[6]
- c) Compose a program to implement the Java AWT Button. [5]

Q4) a) Explain in detail about the event handling mechanism in Java. **[6]**

- b) Explain the concept of Inner Classes and Anonymous Inner Classes with suitable example. [6]

- c) Write a Java program to handle mouse events implementing the `MouseListener` interface. [5]

P.T.O.

- Q5)** a) Explain Flow and Border layout management classes with examples of each. **[6]**
- b) Describe the following: **[6]**
- i) JMenu
 - ii) JMenuItem.
- c) Describe the ArrayList Class in Java Collection framework with suitable example. **[6]**

OR

- Q6)** a) Explain the Java AWT container and components in detail. Define the containers window and panel. **[6]**
- b) Describe the Collection Framework. Explain Map Interface in detail. **[6]**
- c) Compose a Java Swing program to implement a simple Notepad application consisting of JFrame, JMenuBar, JMenu (file, edit, help) and JMenuItem (copy, paste). **[6]**
- Q7)** a) Sketch the JDBC architecture, Using JDBC, explain in detail about the steps to connect to the database in java. **[6]**
- b) Explain in detail about the overview of the JDBC process. **[6]**
- c) Develop a program to insert a table having the fields such as Roll NO., Name, Mobile NO, and also retrieve the same data from the database using JDBC. **[6]**

OR

- Q8)** a) Compare between prepared statement & Callable Statement? **[6]**
- b) Develop a program to insert and retrieve the data from the database using JDBC. **[6]**
- c) Explain how SQL Exception help to determine the cause of the error encounters during execution of JDBC process. **[6]**



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SEAT No. :

PA-1594

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Q1) a) Explain how to generate following components in Swing: **[8]**

- i) Text Fields
- ii) Push Buttons
- iii) Checkbox Group
- iv) Scroll Panes

b) Explain the use of `getDocumentBase( )` and `getCodeBase( )`. **[4]**

c) Explain the mechanism of parameter passing in applets. **[5]**

OR

Q2) a) Write a applet program to display simple login page (title, username, password & submit button) using Swing components. **[8]**

b) What are limitations of AWT? **[4]**

c) What is HTML? Explain any 5 tags used in HTML. **[5]**

Q3) a) What is Event Handling? Explain delegation event model. **[6]**

b) Explain different types of layout managers in java. **[6]**

c) Explain the mechanism of handling keyboard events in java. **[6]**

OR

P.T.O.

- Q4)** a) What are adapter classes? [6]
b) Explain the mechanism of handling mouse events in java. [6]
c) State the difference between AWT & Swing. [6]

- Q5)** a) Define GUI in java. What are components & containers in Java GUI programming. [8]
b) Write a frame program for login page (title, username, password & submit button). Also display welcome message in dialog window. [9]

OR

- Q6)** a) Explain the following: [8]
i) Lists
ii) Queue
iii) Set
iv) Maps
b) Explain hashing mechanism used in collection framework. [4]
c) Draw the hierarchy of collection framework. [5]

- Q7)** a) What are the steps to connect to the database in java? [6]
b) What are the JDBC API components? [6]
c) Write a java program to update a field into table using JDBC. [6]

OR

- Q8)** a) Explain the different types of JDBC statements. [6]
b) What is batch processing? Explain how to execute batch using JDBC. [6]
c) What is the difference between prepared statement and callable statement? Explain. [6]

